

Healthcare Management Analytics

Optimizing Hospital Operations Through Data-Driven Decision Making

1. Company Background

Company Name

MediCare Hospitals Pvt. Ltd.

Industry

Healthcare

Company Overview

MediCare Hospitals Pvt. Ltd. is a leading multi-specialty healthcare provider operating across major cities in India. The hospital network offers comprehensive medical services including Outpatient (OPD), Inpatient (IPD), Emergency Care, Intensive Care (ICU), Diagnostic Services, Pharmacy, Surgery, and Rehabilitation.

Serving over **200,000 patients annually**, the hospital aims to deliver high-quality patient care while continuously improving operational efficiency. With increasing patient volumes, effective management of hospital resources such as doctors, beds, appointments, and medical inventory has become essential. To support strategic decision-making, the hospital intends to leverage data analytics for optimizing operations, improving patient outcomes, and enhancing overall healthcare service quality.

2. Business Problem

Over the past year, MediCare Hospitals has experienced several operational challenges that have affected both efficiency and patient satisfaction. These include:

- Increasing patient admissions resulting in longer waiting times.
- Uneven utilization of hospital beds across departments.
- High patient readmission rates within 30 days of discharge.
- Delays in patient discharge processes.
- Inefficient appointment scheduling causing doctor workload imbalance.
- Rising treatment costs and operational expenses.
- Difficulty in monitoring department-wise performance using existing reports.

The hospital management currently relies on manual reports, making it difficult to identify operational bottlenecks and make timely decisions. A data-driven analytical solution is required to improve operational performance, optimize resource allocation, and enhance patient care.

3. Business Objectives

The primary objectives of this project are:

- Reduce average patient waiting time by **20%**.
- Improve bed occupancy efficiency to the optimal range of **85–90%**.
- Reduce 30-day patient readmission rates.
- Optimize doctor scheduling and workload distribution.
- Monitor treatment costs across departments.
- Improve discharge turnaround time (TAT).
- Increase operational efficiency through data-driven insights.
- Enhance patient satisfaction by improving healthcare services.

4. Stakeholders

Stakeholder	Role	Interest
Chief Executive Officer (CEO)	Executive Leadership	Overall hospital performance
Hospital Administrator	Operations	Resource utilization and efficiency
Chief Medical Officer	Medical Leadership	Clinical performance
Department Heads	Department Management	Department performance
Operations Manager	Hospital Operations	Patient flow management
Doctors	Healthcare Providers	Appointment scheduling and workload
Nursing Staff	Patient Care	Efficient patient management
Pharmacy Manager	Inventory Management	Medicine utilization
Finance Team	Financial Operations	Revenue and treatment costs
Patients	Healthcare Consumers	Better healthcare experience

5. Business KPIs

KPI	Description	Business Purpose
Total Patients	Total number of admitted patients	Measure patient volume
Total Revenue	Total treatment revenue	Financial performance
Average Revenue per Patient	Revenue generated per patient	Revenue efficiency
Average Length of Stay (LOS)	Average hospitalization days	Operational efficiency
Bed Occupancy Rate	Percentage of occupied beds	Resource utilization
Readmission Rate	Patients readmitted within 30 days	Quality of care
Average Waiting Time	Time before consultation	Patient satisfaction
Average Treatment Cost	Average medical expense per patient	Cost monitoring
Department Utilization	Patients treated by each department	Department performance
Doctor Utilization	Patients handled by each doctor	Workforce planning
ICU Occupancy Rate	ICU bed utilization	Critical resource management
Pharmacy Utilization	Medicine consumption	Inventory optimization

6. Domain Knowledge

OPD (Outpatient Department)

Provides medical consultation where patients receive treatment without hospital admission.

IPD (Inpatient Department)

Patients admitted for treatment requiring overnight hospitalization.

Length of Stay (LOS)

Average number of days a patient remains admitted.

Formula

$\text{LOS} = \text{Total Hospital Days} \div \text{Number of Admitted Patients}$

Business Importance

- Indicates operational efficiency.
 - Longer LOS increases treatment costs and reduces bed availability.
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Turnaround Time (TAT)

Time taken to complete a healthcare process.

Examples

- Admission to doctor consultation
- Admission to discharge
- Lab test request to report generation

Lower TAT improves patient satisfaction.

Readmission Rate

Percentage of patients readmitted within 30 days after discharge.

Formula

$\text{Readmitted Patients} \div \text{Total Discharged Patients} \times 100$

High readmission rates often indicate poor treatment outcomes or inadequate follow-up care.

Bed Occupancy Rate

Measures hospital bed utilization.

Formula

$\text{Occupied Beds} \div \text{Total Beds} \times 100$

Ideal occupancy generally ranges between **85% and 90%**.

Patient Flow

The movement of patients through the hospital.

Registration → Appointment → Consultation → Diagnosis → Admission (if required) → Treatment → Pharmacy → Billing → Discharge

Efficient patient flow reduces waiting time and improves hospital operations.

ICD Codes

International Classification of Diseases (ICD) codes are standardized diagnostic codes used worldwide.

Examples

ICD Code	Disease
I10	Hypertension
E11	Type 2 Diabetes
J18	Pneumonia
N18	Chronic Kidney Disease
C50	Breast Cancer

7. Business Requirements

The hospital requires a centralized analytics solution capable of:

- Monitoring patient admissions.
- Tracking doctor performance.
- Measuring department utilization.
- Analyzing treatment costs.
- Monitoring readmission trends.
- Tracking bed occupancy.
- Evaluating appointment efficiency.
- Supporting operational decision-making.

The solution should allow management to monitor hospital performance using interactive dashboards and analytical reports.

8. Dataset Overview

The project consists of multiple relational datasets representing different hospital operations.

Table	Purpose
Patients	Patient demographic and admission details
Doctors	Doctor information

Table	Purpose
Departments	Hospital department information
Appointments	Appointment scheduling records
Billing	Patient billing details
Pharmacy	Medicine inventory records

Expected Dataset Size

- Approximately **8,000–12,000 patient records**
- 6 relational tables
- Around 35–45 attributes
- Primary and Foreign Key relationships

9. Data Dictionary

Patients Table

Column	Description
Patient_ID	Unique patient identifier
Patient_Name	Full name
Age	Patient age
Gender	Male/Female
City	Patient city
Admission_Date	Hospital admission date
Discharge_Date	Discharge date
Department_ID	Department identifier
Doctor_ID	Assigned doctor
Diagnosis_Code	ICD diagnosis code
Treatment_Cost	Total treatment expense
Readmitted	Yes/No
Insurance_Type	Insurance provider

Doctors Table

Column	Description
Doctor_ID	Unique doctor ID
Doctor_Name	Doctor name
Department_ID	Department
Specialization	Medical specialty
Experience_Years	Years of experience

Departments Table

Column	Description
Department_ID	Department identifier
Department_Name	Department name
Total_Beds	Number of available beds
ICU_Beds	ICU bed count

Appointments Table

Column	Description
Appointment_ID	Appointment identifier
Patient_ID	Patient identifier
Doctor_ID	Doctor identifier
Appointment_Date	Appointment date
Waiting_Time	Waiting time (minutes)
Appointment_Status	Completed/Cancelled/No Show

Billing Table

Column	Description
Bill_ID	Billing identifier

Column	Description
Patient_ID	Patient identifier
Total_Bill	Total amount billed
Payment_Method	Cash/Card/Insurance/UPI
Payment_Status	Paid/Pending

Pharmacy Table

Column	Description
Medicine_ID	Medicine identifier
Medicine_Name	Medicine name
Quantity_Issued	Quantity dispensed
Unit_Cost	Cost per unit
Stock_Status	Available/Low Stock/Out of Stock

10. Business Questions

The analysis aims to answer the following business questions:

Patient Analytics

- Which departments receive the highest number of patient admissions?
- Which age groups contribute most to hospital admissions?
- Which cities generate the highest patient inflow?
- Which diseases account for the largest number of admissions?

Doctor Performance

- Which doctors handle the highest patient volume?
- What is the average consultation load per doctor?
- Which departments have an uneven workload distribution?

Hospital Operations

- What is the average patient waiting time across departments?
- Which departments exceed the target Length of Stay (LOS)?
- What is the current bed occupancy rate?

- Which days experience the highest patient admissions?
- What are the busiest OPD hours?

Financial Analysis

- Which departments generate the highest revenue?
- What is the average treatment cost by department?
- Which insurance providers contribute the highest revenue?
- What percentage of bills remain unpaid?

Patient Care Quality

- Which diseases have the highest readmission rates?
- Which patient segments have the longest hospital stays?
- What factors are associated with higher readmission risk?

Inventory & Pharmacy

- Which medicines have the highest monthly demand?
- Which medicines are at risk of stock shortages?
- Which medicines contribute the highest pharmacy costs?